Multiplying Bigger Numbers

Three-digit times two-digit: six pieces, or two rows.

243 × 36: six pieces

	200	40	3
30	6,000	1,200	90
6	1,200	240	18

Hundreds, tens and ones across the top; tens and ones down the side.

6,000 + 1,200 + 90 + 1,200 + 240 + 18 = 8,748

The same in two rows

$$\begin{array}{r} 243 \\ \times \quad 36 \\ \hline 1{,}458 \quad (\times 6) \\ 7{,}290 \quad (\times 30) \\ \hline 8{,}748 \end{array}$$

Row 1: 243 × 6 = 1,458 (the bottom pieces).
Row 2: 243 × 30 = 7,290 (the top pieces).
243 × 30 is 243 × 3 tens = 729 tens.
Add the two rows: 8,748.

Your turn!

312 × 23 = ☐

Multiplying Bigger Numbers: Practice

Split, multiply each piece, add.

Part 1 Hundreds times tens 3 hundreds × 4 tens = 12 thousands.

1) 300 × 40 =	2) 200 × 60 =	3) 500 × 70 =
4) 400 × 30 =	5) 700 × 20 =	6) 600 × 90 =
7) 800 × 50 =	8) 900 × 40 =	9) 200 × 80 =
10) 300 × 70 =	11) 500 × 50 =	12) 400 × 60 =

Multiplying Bigger Numbers: Practice (continued)

Part 2 Six pieces The parts are on the outside. Fill in each piece.

13) **234 × 15 =** ☐

	200	30	4
10	☐	☐	☐
5	☐	☐	☐

☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

14) **162 × 37 =** ☐

	100	60	2
30	☐	☐	☐
7	☐	☐	☐

☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

15) **425 × 23 =** ☐

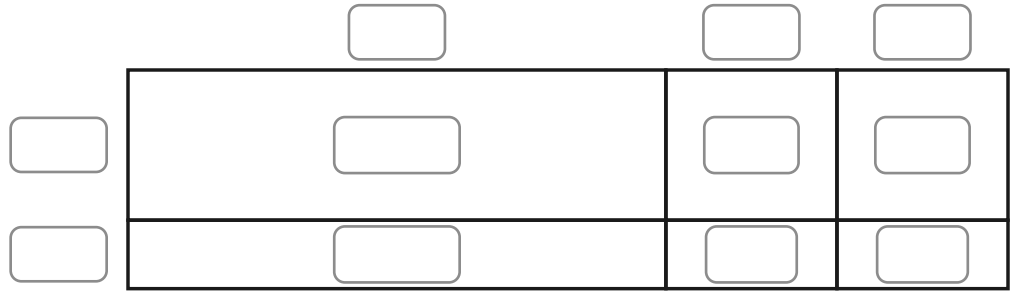
	400	20	5
20	☐	☐	☐
3	☐	☐	☐

☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

Multiplying Bigger Numbers: Practice (continued)

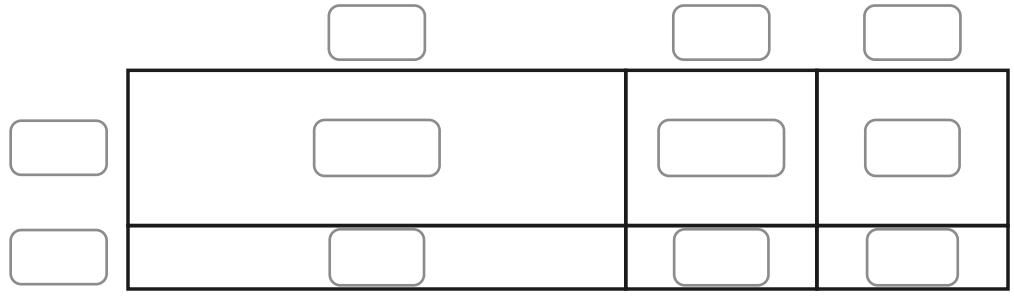
Part 3 Split it yourself Write the parts on the outside, then fill in each piece.

16) **318 × 24 =** ☐



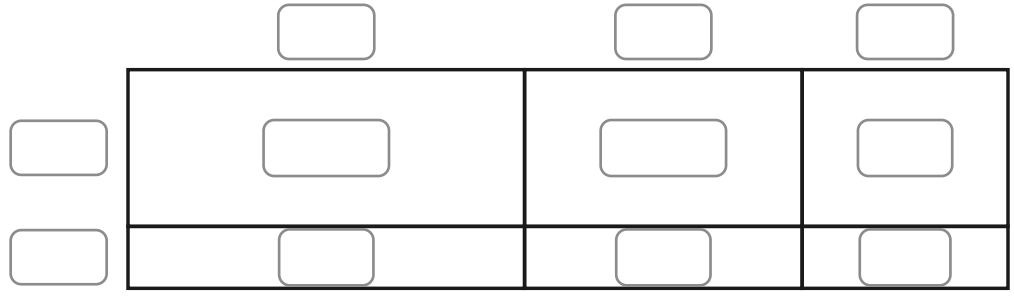
☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

17) **256 × 43 =** ☐



☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

18) **173 × 52 =** ☐



☐ + ☐ + ☐ + ☐ + ☐ + ☐ = ☐

Multiplying Bigger Numbers: Practice (continued)

Part 4 Two rows Row 1: times the ones. Row 2: times the tens. Then add.

19)
$$\begin{array}{r} 214 \\ \times\ 32 \\ \hline \square \quad (\times 2) \\ \square \quad (\times 30) \\ \hline \square \end{array}$$

20)
$$\begin{array}{r} 365 \\ \times\ 24 \\ \hline \square \quad (\times 4) \\ \square \quad (\times 20) \\ \hline \square \end{array}$$

21)
$$\begin{array}{r} 128 \\ \times\ 57 \\ \hline \square \quad (\times 7) \\ \square \quad (\times 50) \\ \hline \square \end{array}$$

22)
$$\begin{array}{r} 432 \\ \times\ 18 \\ \hline \square \quad (\times 8) \\ \square \quad (\times 10) \\ \hline \square \end{array}$$

23)
$$\begin{array}{r} 251 \\ \times\ 46 \\ \hline \square \quad (\times 6) \\ \square \quad (\times 40) \\ \hline \square \end{array}$$

24)
$$\begin{array}{r} 617 \\ \times\ 35 \\ \hline \square \quad (\times 5) \\ \square \quad (\times 30) \\ \hline \square \end{array}$$

25)
$$\begin{array}{r} 183 \\ \times\ 29 \\ \hline \square \quad (\times 9) \\ \square \quad (\times 20) \\ \hline \square \end{array}$$

26)
$$\begin{array}{r} 526 \\ \times\ 14 \\ \hline \square \quad (\times 4) \\ \square \quad (\times 10) \\ \hline \square \end{array}$$

27)
$$\begin{array}{r} 342 \\ \times\ 67 \\ \hline \square \quad (\times 7) \\ \square \quad (\times 60) \\ \hline \square \end{array}$$

Part 5 Multiply No help this time.

28) $326 \times 21 = \square$

29) $147 \times 38 = \square$

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

30) $512 \times 46 =$ ☐

31) $238 \times 17 =$ ☐

32) $403 \times 25 =$ ☐

33) $671 \times 34 =$ ☐

34) $295 \times 42 =$ ☐

35) $184 \times 63 =$ ☐

36) $736 \times 15 =$ ☐

37) $459 \times 27 =$ ☐

38) $128 \times 74 =$ ☐

39) $387 \times 56 =$ ☐

40) $604 \times 39 =$ ☐

41) $215 \times 88 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

42) 548 × 23 = ☐

43) 963 × 12 = ☐

44) 372 × 45 = ☐

45) 819 × 26 = ☐

46) 256 × 91 = ☐

47) 437 × 68 = ☐

Part 6 Four digits times one digit No help this time.

48) 2,134 × 3 = ☐

49) 4,206 × 7 = ☐

50) 1,573 × 4 = ☐

51) 3,058 × 6 = ☐

52) 6,125 × 8 = ☐

53) 2,947 × 5 = ☐

Multiplying Bigger Numbers: Practice (continued)

Part 6 Four digits times one digit (continued)

54) 5,312 × 9 = ____

55) 1,806 × 4 = ____

56) 7,243 × 3 = ____

57) 3,615 × 6 = ____

Part 7 Multiply

58) 672 × 35 = ____

59) 763 × 43 = ____

60) 728 × 41 = ____

61) 163 × 54 = ____

62) 353 × 47 = ____

63) 182 × 39 = ____

64) 672 × 59 = ____

65) 236 × 54 = ____

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

66) $526 \times 16 =$ ☐

67) $162 \times 15 =$ ☐

68) $621 \times 38 =$ ☐

69) $113 \times 29 =$ ☐

70) $873 \times 19 =$ ☐

71) $113 \times 22 =$ ☐

72) $194 \times 12 =$ ☐

73) $664 \times 59 =$ ☐

74) $153 \times 33 =$ ☐

75) $542 \times 49 =$ ☐

76) $371 \times 26 =$ ☐

77) $736 \times 48 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

78) $358 \times 17 =$ ☐

79) $848 \times 59 =$ ☐

80) $237 \times 38 =$ ☐

81) $275 \times 42 =$ ☐

82) $344 \times 58 =$ ☐

83) $355 \times 48 =$ ☐

84) $148 \times 17 =$ ☐

85) $464 \times 12 =$ ☐

86) $497 \times 24 =$ ☐

87) $527 \times 55 =$ ☐

88) $459 \times 14 =$ ☐

89) $604 \times 24 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

90) $182 \times 36 =$ ☐

91) $883 \times 43 =$ ☐

92) $121 \times 13 =$ ☐

93) $165 \times 47 =$ ☐

Part 8 Remember? Problems from earlier packets.

94) $3.6 + 2.75 =$ ☐

95) $8.2 - 4.67 =$ ☐

96) 4.28 to the nearest tenth = ☐

97) 0.45 ☐ 0.405

98) in 3.142 the 4 is worth ☐

99) $2.7 \times 1{,}000 =$ ☐

Multiplying Bigger Numbers: Practice (continued)

Part 9 Word problems Split, multiply, add.

100) A school bus holds 48 kids. How many kids can 125 buses hold?

Number sentence _____

Answer ☐ kids

Draw or show your work

101) A farm packs 236 boxes of eggs. Each box has 12 eggs. How many eggs is that?

Number sentence _____

Answer ☐ eggs

Draw or show your work

102) A plane flies 465 miles each day for 21 days. How far does it fly?

Number sentence _____

Answer ☐ miles

Draw or show your work

103) A box of paper has 1,250 sheets. How many sheets are in 8 boxes?

Number sentence _____

Answer ☐ sheets

Draw or show your work

104) Challenge: A theater has 34 rows of 28 seats. 875 seats are full. How many are empty?

Number sentence _____

Answer ☐ seats

Draw or show your work

Check-In: Multiplying Bigger Numbers

Show what you know. No help this time. Score: _____ / 10

Part 1 Hundreds times tens

1) 300 × 90 = ☐

2) 800 × 90 = ☐

3) 700 × 70 = ☐

Part 2 Two rows

4)

418
× 23

☐ × 3

☐ × 20

☐

5)

152
× 47

☐ × 7

☐ × 40

☐

6)

324
× 26

☐ × 6

☐ × 20

☐

Part 3 Multiply

7) 382 × 19 = ☐

8) 527 × 34 = ☐

9) 243 × 17 = ☐

10) 275 × 36 = ☐